3. $\Gamma(n+1) = n!$, n is a positive integer.

4. $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$



$\Gamma(1)=1$

Example 2.80

Evaluate (i) $\Gamma(6)$ (ii) $\Gamma\left(\frac{7}{2}\right)$

(iii) $\int_0^\infty e^{-2x}x^5dx$ (iv) $\int_0^\infty e^{-x^2}dx$

Solution:

(i) $\Gamma(6) = 5!$

$= 120$

(ii) $\Gamma\left(\frac{7}{2}\right) = \frac{5}{2}\Gamma\left(\frac{5}{2}\right)$

$= \frac{5}{2}\frac{3}{2}\Gamma\left(\frac{3}{2}\right)$

$= \frac{5}{2}\frac{3}{2}\frac{1}{2}\Gamma\left(\frac{1}{2}\right)$

$= \frac{5}{2}\frac{3}{2}\frac{1}{2}\sqrt{\pi} = \frac{15}{8}\sqrt{\pi}$

(iii) we know that

$\int_0^\infty x^n e^{-ax}\,dx = \frac{n!}{a^{n+1}}$

$\therefore \int_0^\infty e^{-2x}x^5\,dx = \frac{5!}{2^{5+1}} = \frac{5!}{2^6}$

(iv) $\Gamma(n) = \int_0^\infty t^{n-1}e^{-t}dt$

Put $t = x^2 \Rightarrow dt = 2xdx$

$\therefore \Gamma(n) = \int_0^\infty e^{-x^2}\left(x^2\right)^{n-1}2x\,dx$

$= \int_0^\infty e^{-x^2}x^{2n-2}2x\,dx$

$= 2\int_0^\infty e^{-x^2}x^{2n-1}\,dx$

Put $n = \frac{1}{2}$, we have

$\Gamma\left(\frac{1}{2}\right) = 2\int_0^\infty e^{-x^2}dx$

$\Rightarrow \sqrt{\pi} = 2\int_0^\infty e^{-x^2}dx$

$\therefore \int_0^\infty e^{-x^2}dx = \frac{\sqrt{\pi}}{2}$

Exercise 2.10

1. Evaluate the following:

(i) $\Gamma(4)$ (ii) $\Gamma\left(\frac{9}{2}\right)$ (iii) $\int_0^\infty e^{-mx}x^6dx$

(iv) $\int_0^\infty e^{-4x}x^4dx$ (v) $\int_0^\infty e^{-\frac{x}{2}}x^5dx$

2. If $f(x) = \begin{cases} x^2e^{-2x}, x \geq 0 \\ 0, \text{ otherwise} \end{cases}$, then evaluate $\int_0^\infty f(x)dx$

2.2.4 Definite integral as the limit of a sum

Let $f(x)$ be a continuous real valued function in $[a, b]$, which is divided into n equal parts of width h, then

$\int_a^b f(x)dx = \lim_{\substack{n\to\infty \\ h\to 0}} h\left[f(a)+f(a+h)+f(a+2h)+.....+f\left(a+(n-1)h\right)\right]$ (or)

$\int_a^b f(x)dx = \lim_{\substack{n\to\infty \\ h\to 0}} \sum_{r=1}^{n} h\, f(a+rh)$,

where $h = \frac{b-a}{n}$

The following results are very useful in evaluating definite integral as the limit of a sum

(i) $1+2+3+.....n = \frac{n(n+1)}{2} = \sum_{r=1}^{n} r$

(ii) $1^2+2^2+3^2+.....n^2 = \frac{n(n+1)(2n+1)}{6} = \sum_{r=1}^{r=n} r^2$

(iii) $1^3+2^3+3^3+.....n^3 = \left[\frac{n(n+1)}{2}\right]^2 = \sum_{r=1}^{r=n} r^3$

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